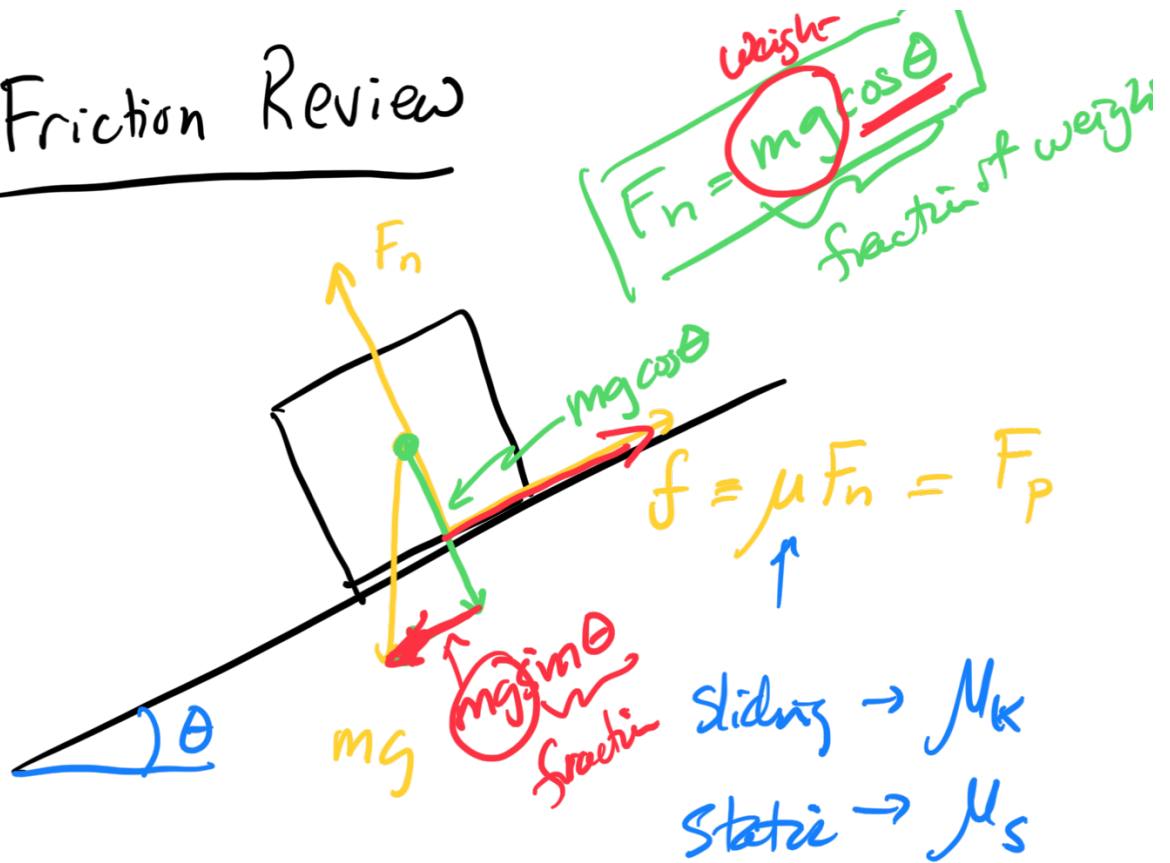


Friction Review

1.



$\rightarrow$ roughness of surface $\rightarrow \underline{\mu_k, \mu_s}$

$\rightarrow F_n \rightarrow \underline{m}, \underline{\text{angle}}$

2.

Weight $\equiv mg$
gravitational force.

$$\begin{aligned}
 3. \quad F_n &= mg \cos \theta \\
 &= (70)(9.8) \cos \theta \\
 F_n &= 686 \cos \theta
 \end{aligned}$$

$$\begin{aligned}
 a) \quad \theta &= 0^\circ & \cos(0^\circ) &= 1 \\
 &(\text{horizontal}) & F_n &= \textcircled{686} \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 b) \quad \theta &= 40^\circ & \cos(40^\circ) &= 0.766 \\
 & & F_n &= \textcircled{525.5} \text{ N}
 \end{aligned}$$

4. static case "sitting"

$$F_p = f = mg \sin \theta$$

$$F_p = 686 \sin \theta$$

$$a) \theta = 0^\circ \quad \sin(0^\circ) = 0$$

$$F_p = 0$$

$$b) \theta = 40^\circ \quad \sin(40^\circ) = 0.6428$$

$$F_p = 686 \cdot 0.6428$$

$$= 440.95 \text{ N}$$

5. "critical angle" ?

↳ pt. at which the object starts to move.

"which two forces are exactly equal"

└ └

$$t_p = t_n$$



$$\cancel{mg \sin \theta} = \cancel{mg \cos \theta}$$

$$\sin \theta = \cos \theta$$

$$\boxed{\theta = 45^\circ}$$

$$\frac{\sin \theta}{\cos \theta} = 1$$

$$\tan \theta = 1$$

$$\theta = \tan^{-1}(1) = 45^\circ$$

6. If the box slides at $\theta = 55^\circ$
 what is the coefficient of friction

$$F_n = mg \cos \theta = 686 \cos 55^\circ = 393.471$$

equal in static case.

$$F_p = f = mg \sin \theta = 561.94$$

$$f \equiv \mu F_n$$

$$561.94 = \mu (393.47)$$

$$\mu = \frac{561.94}{393.47} = 1.428$$

$\mu_s^{\max} \equiv$ maximum coefficient of static friction

7. ✓

Static case.

$$f = \mu F_n$$

$$\cancel{mg \sin \theta} / \cos \theta = \mu (\cancel{mg \cos \theta} / \cancel{\cos \theta})$$

$$\boxed{\mu = \tan \theta}$$

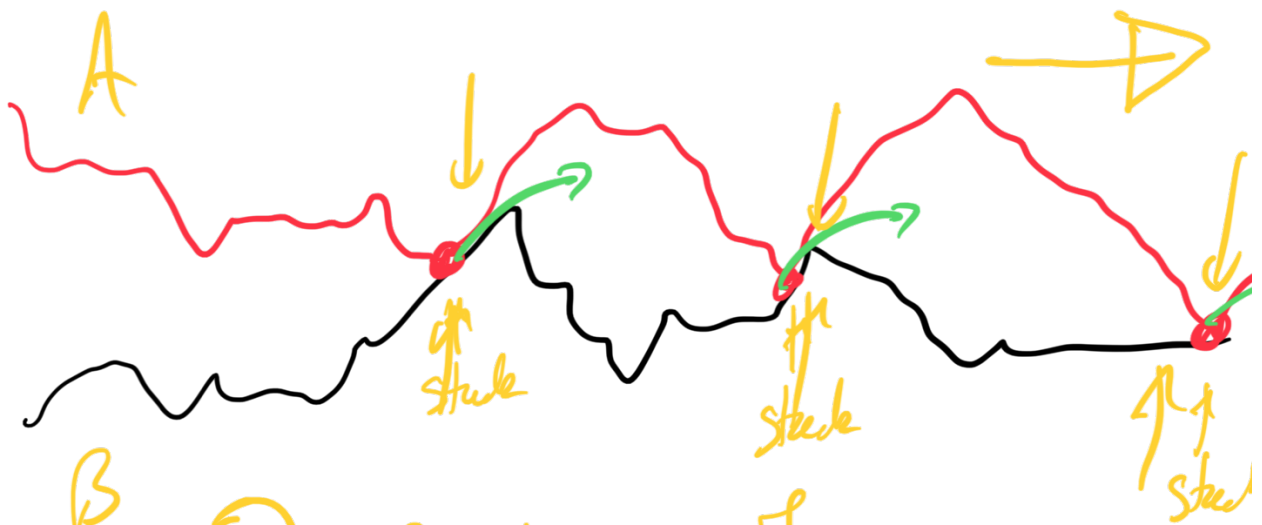
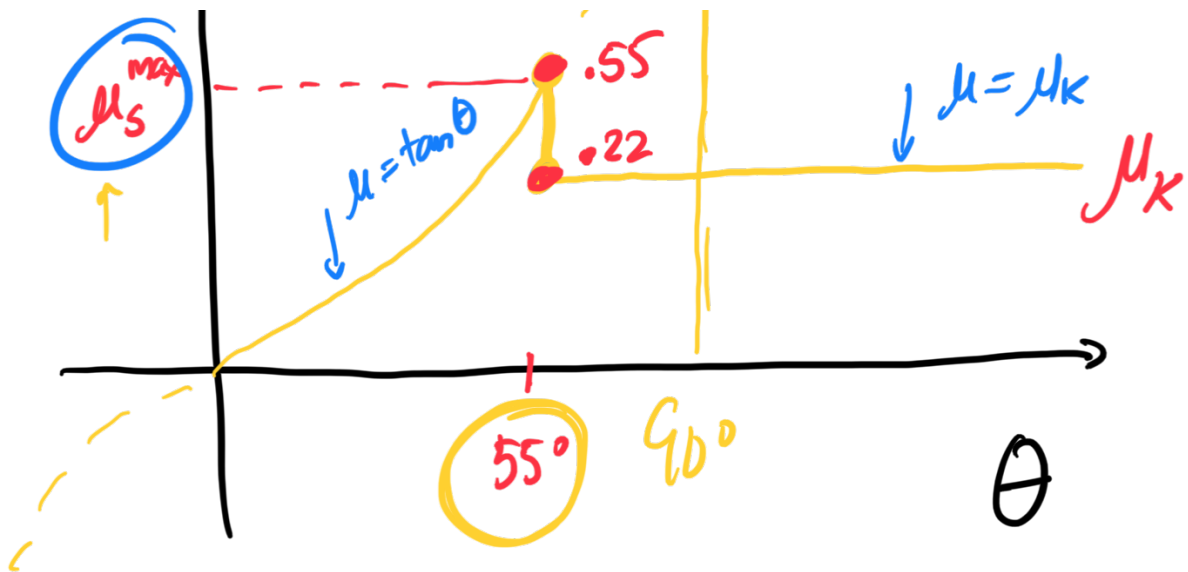


$$\theta_{\text{critical}} = \tan^{-1}(\mu)$$

$\mu \uparrow$

Static

kinetic/sliding



① roughness of surface A, roughness of surface B.

μ is a function of θ.

(2) Molekular ...

(3) Deformationen.



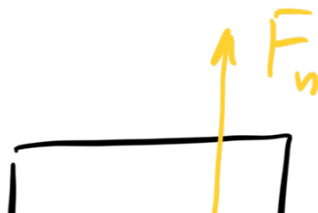
g.

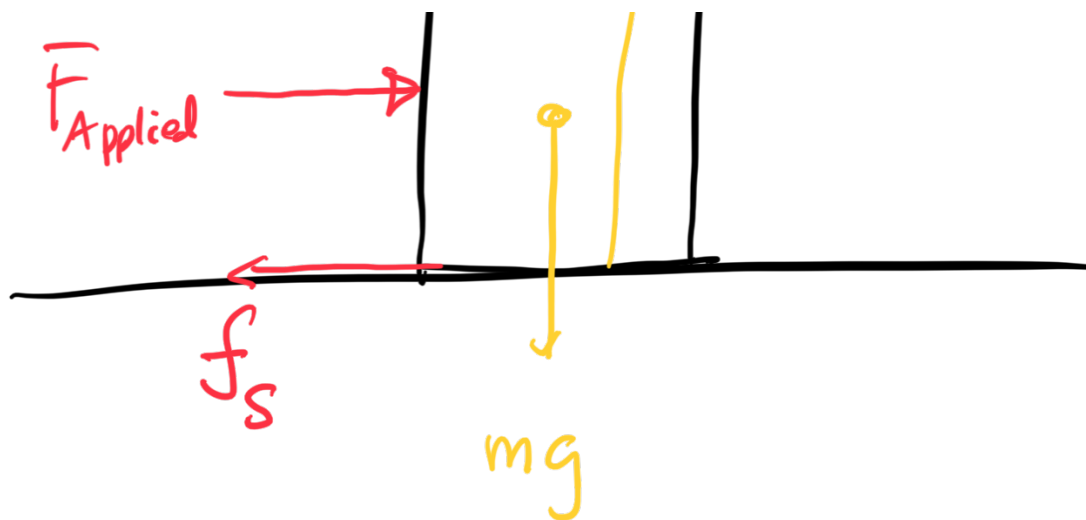
$$m = 14 \text{ kg}$$

$$\mu_s^{\text{max}} = 0.5$$

$$\mu_k = 0.22$$

c)





Static : $\textcircled{F_A} = f_s$
 $F_n = mg$

$$f_s = \mu_s F_n$$

$$f_s = \mu_s mg$$

When it starts to move, barely, $\mu_s = \mu_s^{\text{max}}$

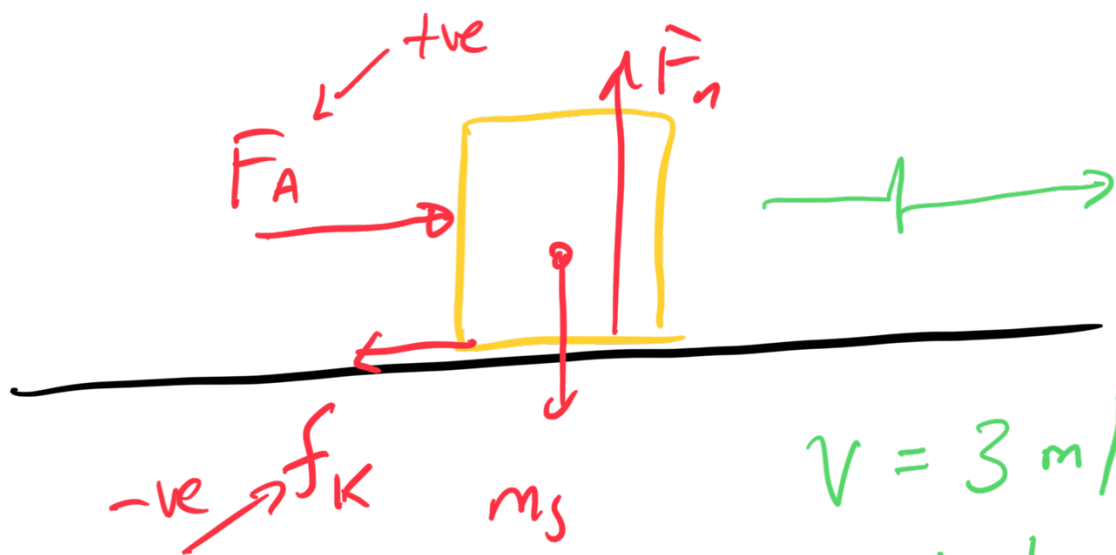
$$f_s = \mu_s^{\text{max}} mg$$

... ..

$$= (0.5)(14)(9.8)$$

$$f_s = \boxed{68.6 \text{ N}} = F_A$$

moving $\rightarrow$ sliding.



$$v = 3 \text{ m/s}$$

$$F_{\text{net}} = ma$$

$$F_A - f_K = 0$$

$$| F_A = f_K |$$

$\rightarrow$ constant

$$\boxed{a = 0}$$

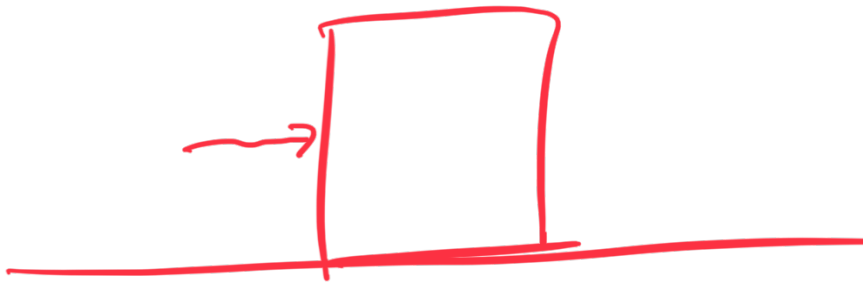
how is the velocity changing?

$$F_n = 14 \text{ N}$$

$$= \mu_k F_n = \mu_k m g$$

$$= (0.22)(14)(9.8)$$

$$= \underline{30.2 \text{ N}}$$



10. $F_A - f_k = m a$

$\uparrow$ $\uparrow$
 60 N $\mu_k m g$
 $= 30.2 \text{ N}$

$$60 - 30.2 = 14 a$$

by 00

$$29.8 = 14a$$

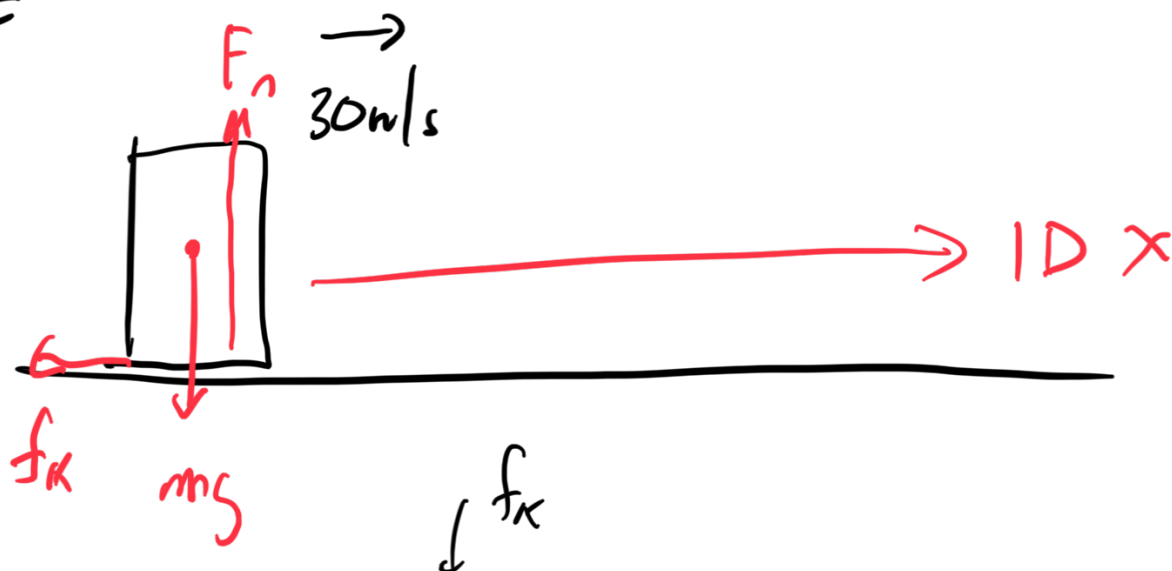
$$a = 29.8 / 14 = 2.13 \text{ m/s}^2$$

b) 240 N

$$(240) - 30.2 = 14a$$

$$a = 14.99 \text{ m/s}^2$$

11.



$$- \mu_k g = a$$

a) ✓

$$\begin{array}{lcl}
 \textcircled{1} & \textcircled{a} & = -\text{MKS} = -\underline{2.156 \text{ m/s}} \\
 & & \uparrow \\
 & & \text{Slowing down.} \\
 \textcircled{2} & \textcircled{V_i} & = \textcircled{30 \text{ m/s}} \quad \sim 60 \text{ mph} \\
 \textcircled{3} & \textcircled{V_f} & = 0 \text{ m/s} \quad (\text{stop}) \\
 \rightarrow & \Delta x & = ?
 \end{array}$$

Kinematics

Motion in 1D with constant acceleration.

$$\begin{array}{lcl}
 \textcircled{1} & V_f & = v_i + at \\
 \textcircled{2} & \Delta x & = v_i t + \frac{1}{2} at^2 \\
 \textcircled{3} & \Delta x & = v_f t - \frac{1}{2} at^2 \\
 \textcircled{4} & \Delta x & = \left(\frac{v_f + v_i}{2} \right) t
 \end{array}$$

$$\textcircled{5} \quad \boxed{V_f^2 = V_i^2 + 2a(\Delta x)}$$

5 unknowns: $V_i, V_f, a, t, \Delta x$

① You will be given/identity $\textcircled{3}$

② Solve for the other two.

$$V_f^2 = V_i^2 + 2a\Delta x$$

$$\Delta x = \frac{V_f^2 - V_i^2}{2a}$$

$$= \frac{0^2 - 30^2}{2(-2.156)}$$

$$= \frac{-900}{-4.312}$$

$$\Delta x = 208.7 \text{ m}$$

is it in?

How long does it take!

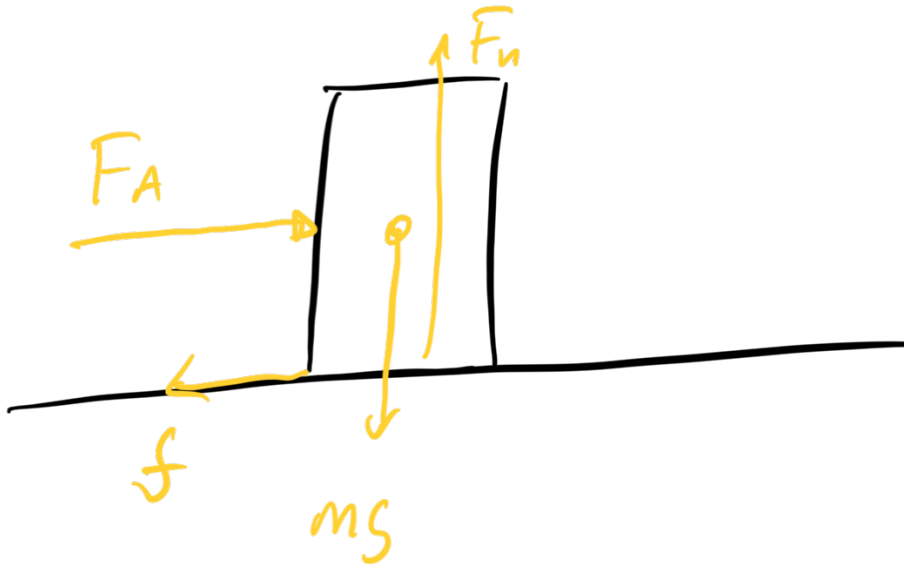
$$t = ?$$

$$V_f = V_i + a t$$

$$t = \frac{V_f - V_i}{a} = \frac{0 - 30}{-2.156}$$

$$= 13.9 \text{ s}$$

12.



$$F_n = mg \quad \leftarrow \text{double the mass, double } F_n$$

$$f = \mu F_n = \mu mg \quad \text{double } f.$$

